

FPGA CODEC comparisons
Advances in RNA Appendix A

James Brakefield

directory name: BM/NS H/_ generic format & size	memory format	ALU bits	ALU exp	ALU signif icand	zero NaR stat	ALU roun ding	mem bits	max exp bits	fract size max	fract size min	total binad es	decad es / 2	mem BMF?	mem HUB?	directi on	FPGA device	LUT cnt	DSP cnt	delay ns	fpga tool	fpga optimization	run date	test bench	valid ated	RTL author VHDL-2008	comments	
The four memory format types																											
NSHld_takum16	Takum	24	9	13	1		16	9	11	4	512	77.1	NFSF	HUB	load	zynq uscale+										Laslo Hunhold	standard linear takum
NSHld_bposit16	Posit sB=5	24	9	13	1		16	8	8	0	400	60.2	NFSF	HUB	load	zynq uscale+										Florent de Dinec	Sb field always present, e.g. limited regime field
NSHld_wmesa16	wide Mesa	24	9	13	1		16	6	9	0	352	53.0	NFSF	HUB	load	zynq uscale+										Jim Brakefield	wide mesa
NSHld_pedestal16	pedestal	24	9	13	1		16	8	7	7	254	38.2	NFSF	HUB	load	zynq uscale+										Jim Brakefield	min implementation of Gustafson criteria
NSHld_nmesa16	narrow Mesa	24	9	13	1		16	6	9	0	52	7.8	NFSF	HUB	load	zynq uscale+										Jim Brakefield	P3109 mesa
Pedestal with HUB encoding for zero, NaR, +/- 1.0, no sub or super normals																											
NS_ld_pedestal16	pedestal8_16	24	9	13	1	RTE	16	8	7	7	254	38.2	NFSF		load	zynq uscale+	3		0.940	vivado 26.1	area remap	06/04/26				Jim Brakefield	for comparison to Bfloat16
NSHld_pedestal16	pedestal8_16	24	9	13	1	ITO	16	8	7	7	254	38.2	NFSF	HUB	load	zynq uscale+	3		0.810	vivado 26.1	area remap	06/04/26				Jim Brakefield	decpdz zero, NaR and +/- 1.0
NS_st_pedestal16	pedestal8_16	24	9	13	1	RTE	16	8	7	7	254	38.2	NFSF		store	zynq uscale+	21		1.120	vivado 26.1	area remap	06/04/26				Jim Brakefield	round to nearest, ties to even
NSHst_pedestal16	pedestal8_16	24	9	13	1	ITO	16	8	7	7	254	38.2	NFSF	HUB	store	zynq uscale+	8		0.750	vivado 26.1	area remap	05/28/26				Jim Brakefield	encode zero, NaR and +/- 1.0
Wide taper Mesa with HUB encoding for zero, NaR, +/- 1.0																											
BM_ld_wMesa6_16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	BMF		load	zynq uscale+				vivado 26.1	area remap					Jim Brakefield	wide mesa, nearly Takum exponent range
BMHld_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_ld_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	NFSF		load	zynq uscale+	21	0	1.355	vivado 26.1	area remap	08/25/26			Jim Brakefield		
NSHld_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	NFSF	HUB	load	zynq uscale+	16	0	1.341	vivado 26.1	area remap	08/25/26			Jim Brakefield		
BM_st_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	BMF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
BMHst_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_st_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NSHst_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
Narrow taper Mesa with HUB encoding for zero, NaR, +/- 1.0																											
BM_ld_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	BMF		load	zynq uscale+				vivado 26.1	area remap					Jim Brakefield	for comparison to IEEE Float16
BMHld_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_ld_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	NFSF		load	zynq uscale+	34	0	0.956	vivado 26.1	area remap	08/22/26			Jim Brakefield		
NSHld_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	NFSF	HUB	load	zynq uscale+	24	0	1.404	vivado 26.1	area remap	08/22/26			Jim Brakefield		
BM_st_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	BMF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
BMHst_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_st_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	NFSF		store	zynq uscale+	242	0	4.850	vivado 26.1	area remap				Jim Brakefield	incorrect round ties to even	
NSHst_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	NFSF	HUB	store	zynq uscale+	32	0	1.362	vivado 26.1	area remap	09/04/26			Jim Brakefield		
Linear Takum with HUB encoding for zero, NaR, +/- 1.0																											
BM_ld_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	BMF		load	KntexUltrScal+	39		3.652	vivado 24.1	default				Laslo Hunhold	linear and logarithmic varinats defined	
BMHld_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield	from paper's TeX:Dsgn & Imp takum arithmetic codec	
NS_ld_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	NFSF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NSHld_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	NFSF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
BM_st_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	BMF		store	KntexUltrScal+	66		4.180	vivado 24.1	default				Laslo Hunhold	from paper's TeX:Dsgn & Imp takum arithmetic codec	
BMHst_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_st_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NSHst_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
Wide Posit (eS=5) with HUB encoding of zero, NaR, +/- 1.0																											
BM_ld_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	BMF		load	KntexUltrScal+	57		3.911	vivado 24.1	default				Florent de Dinec	from paper TeX:Dsgn & Imp takum arithmetic codec. FloPoCo	
BM_st_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	BMF		store	KntexUltrScal+	56		4.349	vivado 24.1	default				Florent de Dinec	from paper TeX:Dsgn & Imp takum arithmetic codec. FloPoCo	
BMHld_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
BMHst_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_ld_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	NFSF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NS_st_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NSHld_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	NFSF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
NSHst_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield		
To Establish Suitable Context for the CODEC results																											
add/subtract		24	9	13	1	ITO										zynq uscale+	300-400			vivado 26.1	area remap					AMD IP	codecs used on memory load/store, not within ALU?
f12_e9_mult_zu3	multiply	24	9	13	1	ITO	NA	10	13	NA	512	77.1	NA	NA	NA	zynq uscale+	43	1	2.700	vivado 26.1	area remap	07/27/26			Jim Brakefield	pipelined, no speed optimization, round ties to even?	
f12_e9_mult_zu3	multiply	24	9	13	1	ITO	NA	10	13	NA	512	77.1	NA	NA	NA	zynq uscale+	34	1	2.920	vivado 26.1	area remap	07/28/26			Jim Brakefield	saturation, extra exp bit, hidden fract bit	
divide combinatorial		24	9	13	1	ITO										zynq uscale+				vivado 26.1	area remap				AMD IP	ibid. clk of 4.0ns instead of 3.0ns	
divide via reciprocal table		24	9	13	1	ITO										zynq uscale+				vivado 26.1	area remap				Jim Brakefield	uses block RAM for reciprical lookup	

Round Inexact To Odd (IEEE P754 Round to one)

ITO

Round ties to even

RTE

NFSF: No Fraction Shifter Format

NS

BMF: Binary Monotonic Format

BM

HUB: Half Unit Bias AKA Implicit half bit set format

H

Binade: Approximate binary mem exp range: total binary exponent range

For this study, all binades have atleast one entry

1/2 Decade: Approximate half decimal range, e.g. +/- total decimal exponent range

On HUB load implicit HUB bit is inserted into the ALU fraction

On store and non-HUB the ALU fraction is rounded ties to even

In general on load and store the resulting value is true to the source value, rounded as appropriate

Timing results via enclosing combinatorial logic with input and output registers

The pedestal AKA flat roof format supports zero, NaR and saturation and no sub-normals

Directory naming: BM/NS, H/_ , LD/ST, _name-size

wide Mesa16: six bit exponent with 32 of the 64 exponents used on the tapers (sub and super-normals)

Fraction size of zero means only the hidden leading one bit present

Saturation can occur on memory stores due to exponent size reduction or if round ties to even increases the exponent